

Fourier Series for Continuous-time Signals

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{T})t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jk(\frac{2\pi}{T})t} dt$$

$$= \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jk\omega_0 t} dt$$

Fourier Transform for Continuous-time Signals

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Fourier Series for Discrete-time Signals

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(\frac{2\pi}{N})n} = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(\frac{2\pi}{N})n}$$

$$= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n}$$

Fourier Transform for Discrete-time Signals

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

Discrete Fourier Transform

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(\frac{2\pi}{N})nk}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(\frac{2\pi}{N})nk}$$

Fourier Transform Properties & Pairs

- FT

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$$

$$e^{j\omega_0 t} x(t) \leftrightarrow X(j(\omega - \omega_0))$$

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

$$x(t) * y(t) \leftrightarrow X(j\omega) Y(j\omega)$$

$$x(t)y(t) \leftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega)$$

$$\frac{d}{dt} x(t) \leftrightarrow j\omega X(j\omega)$$

$$t x(t) \leftrightarrow j \frac{d}{d\omega} X(j\omega)$$

$$\int_{-\infty}^t x(t) dt \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega)$$

$$\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \leftrightarrow 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0)$$

$$e^{j\omega_0 t} \leftrightarrow 2\pi \delta(\omega - \omega_0)$$

$$\delta(t) \leftrightarrow 1$$

$$1 \leftrightarrow 2\pi \delta(\omega)$$

$$u(t) \leftrightarrow \frac{1}{j\omega} + \pi \delta(\omega)$$

- DTFT

$$x[n - n_0] \leftrightarrow e^{-jn_0\omega} X(e^{j\omega})$$

$$e^{jn_0\omega} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$$

$$x_{(k)}[n] \begin{cases} x\left[\frac{n}{k}\right] & \text{if } n = \text{multiply of } k \\ 0 & \text{if } n \neq \text{multiply of } k \end{cases} \leftrightarrow X(e^{jk\omega})$$

$$x[n] * y[n] \leftrightarrow X(e^{j\omega}) Y(e^{j\omega})$$

$$x[n]y[n] \leftrightarrow \frac{1}{2\pi} X(e^{j\omega}) * Y(e^{j\omega})$$

$$x[n] - x[n-1] \leftrightarrow (1 - e^{-j\omega}) X(e^{j\omega})$$

$$y[n] = \sum_{k=-\infty}^n x[k]$$

$$\leftrightarrow \frac{1}{(1 - e^{-j\omega})} X(e^{j\omega})$$

$$+ \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

$$n x[n] \leftrightarrow j \frac{d}{d\omega} X(e^{j\omega})$$

$$\sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{N})t} \leftrightarrow 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - \frac{2\pi k}{N})$$

$$e^{jn\omega_0} \leftrightarrow 2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi l)$$

$$\delta[n] \leftrightarrow 1$$

$$1 \leftrightarrow 2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l)$$

$$u[n] \leftrightarrow \frac{1}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

$$a^n u[n] \text{ where } |a| < 1 \leftrightarrow \frac{1}{1 - ae^{-j\omega}}$$

- DFT (circular shift, periodic convolution)

$$x[n + n_0] \leftrightarrow e^{j(\frac{2\pi}{N})n_0 k} X[k]$$

$$X[k + k_0] \leftrightarrow e^{-j(\frac{2\pi}{N})k_0 n} x[n]$$

$$x[n] * h[n] \leftrightarrow X[k]H[k]$$

Discrete-time Convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Continuous-time Convolution

$$y(t) = \int_{\tau=-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

Odd-even functions

Even signal: $x(-t) = x(t)$

Odd signal: $x(-t) = -x(t)$

A signal $x(t)$ has

Even part: $x_e(t) = \frac{1}{2}(x(t) + x(-t))$

Odd part: $x_o(t) = \frac{1}{2}(x(t) - x(-t))$

Energy – Power for Continuous-time Signals

Total energy $E = \lim_{L \rightarrow \infty} \int_{-L}^L |x(t)|^2 dt$

Average power $P = \lim_{L \rightarrow \infty} (\frac{1}{2L} \int_{-L}^L |x(t)|^2 dt)$

If $0 < E < \infty$, $x(t)$ is an energy signal with zero power

If $0 < P < \infty$, $x(t)$ is a power signal with infinite energy

Energy – Power for Discrete-time Signals

Total energy $E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2$

Average power $P = \lim_{N \rightarrow \infty} (\frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2)$

Unit Impulse properties

$$\int_{t_1}^{t_2} x(t)\delta(t-t_0)dt = \begin{cases} x(t_0), & t_1 < t_0 < t_2 \\ 0, & \text{otherwise} \end{cases}$$

$$x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$$

$$\delta(at+b) = \frac{1}{|a|}\delta\left(t+\frac{b}{a}\right)$$

Integration by parts

$$\int u dv = uv - \int v du$$

Geometric Series

$$\sum_{k=0}^N a_1 r^k = a_1 \frac{1 - r^{N+1}}{1 - r}$$

$$\sum_{k=0}^{\infty} a_1 r^k = \frac{a_1}{1 - r}$$

where $r \neq 1$, a_1 is the first term, r is the ratio

Complex numbers

$$z = a + bj = \text{Re}\{z\} + j\text{Im}\{z\} = |z|e^{j\angle z}$$

$$\text{Magnitude: } |z| = |a + bj| = \sqrt{a^2 + b^2}$$

$$\text{Phase: } \angle z = \tan^{-1}\left(\frac{b}{a}\right)$$

Euler's formula

$$e^{j\theta} = \cos(\theta) + j\sin(\theta)$$

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Trigonometric Formula

$$\sin(A \pm B) = \sin(A) \cos(B) \pm \cos(A) \sin(B)$$

$$\cos(A \mp B) = \cos(A) \cos(B) \pm \sin(A) \sin(B)$$

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$\sin^2 A + \cos^2 A = 1$$

$$1 - \cos 2A = 2\sin^2 A$$

$$\frac{d}{dx}(\sin x) = \cos(x)$$

$$\frac{d}{dx}(\cos x) = -\sin(x)$$

$$\frac{d}{dx}(\tan x) = \sec^2(x)$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\cot x) = -\csc^2(x)$$

Sampling

$f_p = |f - kf_s|$ where k is the rounded value of $\frac{f}{f_s}$,

f_p is the perceived frequency, f is the frequency of the signal and f_s is the sampling rate.